

House Price Prediction – Week 1 Project Summary

Objective

Build a machine learning model to predict house prices using property features such as area, bedrooms, bathrooms, stories, parking, furnishing status, and amenities.

Dataset

Housing.csv with 545 records and 13 property-related features. Target variable: Price.

Data Preprocessing

Checked for missing values and duplicates, encoded categorical variables, applied one-hot encoding, and split the dataset into training and testing sets.

Models Implemented

1. Linear Regression
2. Random Forest Regressor

Results

Linear Regression: MAE = 970,043 | RMSE = 1,324,507 | $R^2 = 0.653$

Random Forest: MAE = 1,014,947 | RMSE = 1,399,769 | $R^2 = 0.612$

Key Findings

Area, bathrooms, stories, air conditioning, and preferred area were the most important features influencing house prices. Linear Regression outperformed Random Forest and explained about 65% of the variation in house prices.

Conclusion

The project successfully developed a house price prediction system. Linear Regression provided the best predictive performance and identified the most influential factors affecting property values.